GITAM Deemed to be University

Information Technology | Course Code: IT303

Cyber Security & Cryptography - Comprehensive Syllabus Lecture Notes

Syllabus Focus: IT303 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Cyber Security & Cryptography. Includes detai

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Symmetric Cryptography & Block Ciphers

" Syllabus Scope: Feistel cipher model, DES, Triple-DES, Advanced Encryption Standard AES rounds, Modes o

Unit 2: Asymmetric Cryptography & Number Theory

" Syllabus Scope: Modular arithmetic, Extended Euclidean algorithm, RSA cryptosystem, Diffie-Hellman key

IT303: Cyber Security & Cryptography - Advanced Topics & Assessment

Academic Track: Information Technology - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Cryptographic Hash & Digital Signatures

" Hash functions, SHA-256, HMAC construction, Digital signatures DSA, ECDSA, Public Key Infrastructure PKI, X.

Unit 4: Network Security Protocols

" TLS 1.3 handshake, IPsec AH and ESP, Firewalls Stateful packet inspection, Intrusion Detection Systems IDS,

Unit 5: Web Security & Cyber Defenses

" OWASP Top 10 vulnerabilities, SQL injection, XSS, Buffer overflow exploits, Threat modeling, Zero-Trust netw

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Trace the mathematical steps of RSA algorithm: calculate public and private keys for primes p=17, q=11
2. Describe the Diffie-Hellman key exchange protocol and analyze its vulnerability to Man-In-The-Middle M
3. Formulate the four transformations applied in each round of the Advanced Encryption Standard AES ciphe

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* William Stallings - Cryptography and Network Security: Principles and Practice Pearson 7th Ed
* Bruce Schneier - Applied Cryptography: Protocols, Algorithms, and Source Code in C Wiley

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